

Communication Complexity: Communication Lower Bounds For Multi-Party Graph Traversal

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Robots

Flights

Why Study Communication Complexity

Communication complexity studies the communication requirements to solve problems where inputs are distributed among at least 2 parties with unbounded computing power. The 2 party problem is typically stated as the following:

There are two players, Alice and Bob who each have a n -bit string, a and b that may or may not be the same. The goal is to compute a function, $f(a, b)$ that depends on both a and b .

Under the two party problem, it is always possible to compute $f(a, b)$ by sending all of a to b , effectively reducing problem to a single party. This approach of sending the entire input from one party to another is what we consider the *trivial solution* and is *maximally hard*. We consider this *maximally hard* because it has a communication complexity of $\theta(n)$ bits as all inputs must be sent from a to b . One of the most common examples of a *maximally hard* problem is 2 Player Equality.

Communication Complexity of 2 Player Equality

TODO Write the equality proof

By studying communication complexity, we seek to find creative ways to achieve non-trivial bounds on multiparty problems or embed proven bounds in novel problems.

Note: Communication complexity is fundamentally disjoint from time complexity as it studies the communication under the assumption of unlimited computing power.

The Graph Traversal Problem

We consider the case where we have a known graph with n nodes and an unknown number of edges, and two parties, Alice and Bob, with connected subgraphs $a, b \in \{0, 1\}^n$. Our goal is to determine whether or not, given a starting node v in a and ending node w in b , we can find a path from v to w such that every edge traversed exists within either a or b .

TODO Add figures for the graph traversal here.

More formally, we define our problem, denoted as $TRAV : \{0, 1\}^n \times \{0, 1\}^n \rightarrow \{0, 1\}$, by $TRAV(x, y) = 1$ if there exists a path from v to u such that all edges traversed belong to either x or y .

Deterministic Protocols and Maximally Hard

Overview. Building on the two-party model defined previously, in this section we will formalize deterministic protocols and maximally hard functions. We then prove that Equality is deterministically maximally hard and embed it into TRAV to show that TRAV is also maximally hard.

Deterministic Protocol. A fixed, non-randomness communication procedure that outputs $f(x, y)$ correctly for every input pair. At each step, the next message is determined solely by the party's previous input and the transcript so far.

Maximally Hard. A function f is maximally hard if $D(f) = \Theta(n)$

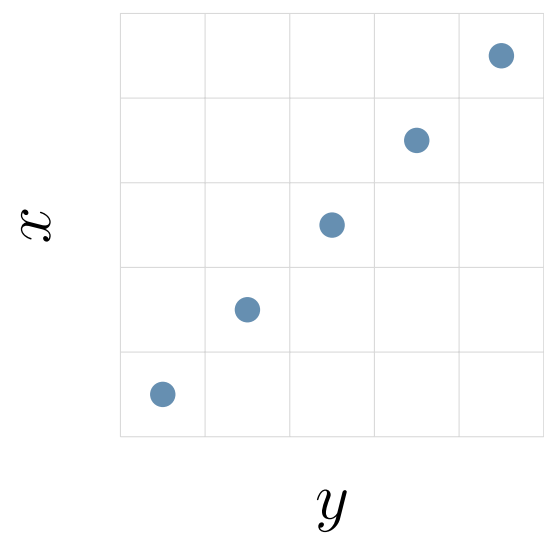
In the worst case, one party can always solve the problem by sending all n of their input bits to the other party. Therefore, linear communication is the largest possible asymptotic complexity. Any function requiring $\Omega(n)$ bits is maximally hard.

Equality is Deterministically Maximally Hard

Equality Problem: Do Alice and Bob hold the same n -bit string?

$$EQ(x, y) = \begin{cases} 1 & x = y \\ 0 & x \neq y \end{cases}$$

We visualize EQ using its communication matrix: rows are Alice's inputs, columns are Bob's inputs, and a dot marks where $EQ(x, y) = 1$.



All 1-outputs lie on the diagonal, since equality holds iff $x = y$. Every off-diagonal entry is 0.

A deterministic protocol must partition this matrix into *monochromatic rectangles*, meaning submatrices where all entries have the same output (all 0's or all 1's).

Because the 1's lie only on the diagonal, no two distinct diagonal entries can share such a rectangle without including a 0. Thus each diagonal 1 must be isolated.

There are 2^n diagonal entries, so $D(EQ) \geq n$. Thus this must mean Equality is maximally hard.

Embedding Equality into TRAV

To prove that TRAV is maximally hard, we reduce Equality to TRAV.

For each bit position i , we construct a gadget that depends on Alice's bit x_i and Bob's bit y_i .

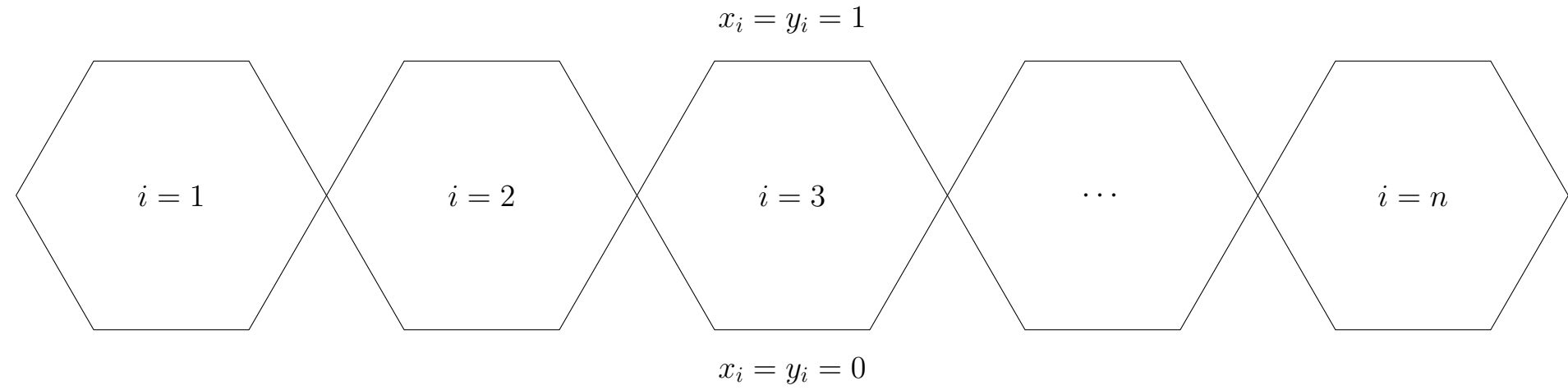


Figure 1. Chain of Equality Gadgets

In our gadget, the path is only traversable at index i if x_i and y_i are both 0 or 1. In other words each gadget is traversable if and only if $x_i = y_i$.

We connect all gadgets in sequence. Therefore, a path exists from the first gadget to the last if and only if $x_i = y_i$ for all i , i.e., if and only if $x = y$.

Thus,

$$EQ \leq TRAV.$$

Since EQ is deterministically maximally hard, TRAV must also require $\Omega(n)$ communication.

Therefore, TRAV is deterministically maximally hard.

Random

Disjoint

Disjoint Gadget